

Coherence as Local Spacetime Geometry

A Hypothesis: Phase-Locked Coupled-Oscillator Systems May Generate Local Spacetime-Curvature Effects Analogous to Gravitational Time Dilation

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Abstract

We propose, as a falsifiable hypothesis, that biological and physical systems exhibiting strong coherent phase-lock among their coupled-oscillator components generate local spacetime-curvature effects structurally analogous to gravitational time dilation, producing measurable alterations of subjective time-density within the coherent region and (potentially) measurable acoustic-frequency warp in signals crossing the coherent region's boundary. The hypothesis treats the well-documented general-relativistic time-dilation effect of mass-energy concentrations and the proposed coherence-induced subjective time-density effect as two articulations of a more general principle: that local concentrations of phase-locked structure curve local spacetime geometry. Specific predictions are advanced regarding (a) the time-dilation behavior of subjects in deep coherent states (meditation, flow, certain pharmacologically-induced states), (b) the propagation of acoustic signals through the spatial extent of such states' biofields, (c) the metabolic cost of sustaining such states, and (d) the relation between coherent-bubble subjective time-density and the dissipative-structure equation governing biological aging. We are explicit that this is a hypothesis paper, not a result paper. The supporting evidence is suggestive but not definitive. The principal value of the paper is to make the hypothesis specific enough to test.

1. Introduction

This paper advances a speculative hypothesis. It does not present new experimental results. It identifies a structural parallelism between two well-documented phenomena and proposes that the parallelism reflects a deeper underlying mechanism. The proposed mechanism is testable, and we provide specific predictions whose confirmation or refutation would adjudicate the hypothesis.

1.1 Two well-documented phenomena

Gravitational time dilation. General relativity predicts, and experiments confirm to high precision, that clocks deeper in a gravitational potential run slower, as measured by clocks at higher gravitational potentials. The effect is not an artifact of measurement; it is a real difference in the rate at which time elapses, integrated by physical processes throughout the gravity well. The Hafele-Keating experiment (1971), Pound-Rebka (1959), GPS satellite corrections, and routine atomic-clock comparisons all confirm the effect to nine or more digits of precision. Time dilation in this regime is unambiguously a real physical phenomenon arising from the curvature of spacetime by mass-energy.

Subjective time dilation in coherent states. A separate body of literature documents that subjects in altered states of consciousness — deep meditation, flow states, near-death experiences, certain pharmacologically-induced states (DMT, psilocybin, ketamine), acute survival situations — frequently report dramatic alterations of subjective time-rate. Reports include: extended time experienced within a brief clock-time window; "everything moving in slow motion"; "lifetimes lived in seconds"; "no sense of time passing." These reports are reliable across diverse subjects, cultures, and contexts. They have been documented in clinical, contemplative, and survival literatures for over a century.

The standard explanation for subjective time dilation in altered states is *neurological*: the integration window of the brain (gated by cardiac, respiratory, and alpha-rhythm coupled-oscillator activity) shifts during these states, processing more or fewer perceptual ticks per external second. Time dilation is then a property of how the nervous system samples and integrates external reality, not of external reality itself.

We propose that this standard neurological explanation may be incomplete. Specifically, we propose that coherent phase-lock states may generate genuine local alterations of spacetime geometry, of which the subjective time-density alteration is the local-observer report.

1.2 The structural parallelism

Both gravitational time dilation and coherent-state time dilation share a striking structural pattern:

Feature	Gravitational case	Coherent-state case	
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Local clock rate alteration	Yes (slowed in well)	Yes (subjectively altered)	
Source	Mass-energy concentration	Phase-lock concentration	

Has a measurable spatial extent	Yes (gravitational radius)	Reported anecdotally
Affects signals crossing the boundary	Yes (gravitational redshift)	Reported anecdotally
Difference between local and external observers	Yes	Yes
Reversal not symmetric (twin-paradox-like)	Yes	Reported (lived-density retained)

The parallelism is suggestive but not proof. The standard neurological account explains the subjective effects without invoking spacetime alteration. The parallelism could be coincidental — both effects might happen to share these features without sharing an underlying mechanism.

We propose that the parallelism is not coincidental, and that there is a more general principle of which both phenomena are special cases. The general principle: **local concentrations of coherent phase-locked structure (whether mass-energy concentrations curving spacetime by general relativity, or coherent biological coupled-oscillator concentrations curving spacetime by an analogous but distinct mechanism) alter local spacetime geometry, producing measurable time-rate effects.**

This is a strong claim. We label it explicitly as a hypothesis and devote the rest of this paper to making it specific enough to test.

2. The Hypothesis

2.1 Statement

Coherence-curvature hypothesis. Concentrations of phase-locked coupled-oscillator coherence — whether mass-energy concentrations or biological-electromagnetic coherent-bubble concentrations — produce local alterations of spacetime metric. The alterations follow a common functional form: increased coherence concentration increases the local rate of subjective integration of physical processes per unit external time, manifesting as subjective time-density alteration in observers within the coherent region and as redshift-like frequency warp in signals propagating from the coherent region to external observers.

The mechanism for the gravitational case is well-established (Einstein field equations). The mechanism for the biological-coherent case is hypothesized but not derived. The hypothesis is that a derivation exists, in which mass-energy concentration and coherent-phase-lock concentration appear as two limiting cases of a more general spacetime-curvature equation.

2.2 What the hypothesis is not

To prevent overreach: the hypothesis is *not* that consciousness itself curves spacetime. Consciousness is not invoked. The proposed source of curvature is the coherent phase-lock structure of the underlying biological coupled-oscillator system, which is a physical (though biological) phenomenon, not a mental one.

The hypothesis is *not* that coherent meditators have measurable mass changes. Mass-energy and phase-lock-coherence are proposed as distinct sources of the same general curvature effect, not as the same thing. A meditator does not become heavier; they generate a curvature contribution from a different mechanism.

The hypothesis is *not* a claim that established physics is wrong. General relativity is presumed correct in its domain. The hypothesis is that there may be an additional mechanism for spacetime curvature, not that the existing mechanism is in error.

2.3 The carrier-configuration framing

The hypothesis is informed by, and consistent with, the geometric arrow framework developed in the companion paper *Geometric Time Directionality*. In that framework, non-integrable coupled-oscillator systems generate carrier waves with strict structural directionality. Coherent phase-lock concentrations may be understood as local regions of the carrier with elevated phase-locking density, in the same way that gravitational masses are local regions of mass-energy density. The carrier framework provides a natural setting in which both kinds of curvature might be expressed as local structure of the same underlying coupled-oscillator field.

3. Empirical Considerations

3.1 Reported anecdotal evidence

Several classes of report are consistent with the hypothesis but do not by themselves confirm it. We list them with care to distinguish anecdote from data:

Class A — Subjective time-dilation reports in coherent states. Extensively documented across meditation traditions (Goleman 1977, Tart 1972), flow-state research (Csikszentmihalyi 1990), near-death experience reports (van Lommel 2001), psychedelic-state reports (Strassman 2001), acute survival situations (Noyes & Kletti 1976). The reports are consistent across thousands of subjects in multiple traditions and contexts. The reports are *consistent with* the hypothesis. They do not by themselves require it; the standard neurological account also predicts them.

Class B — Reports of acoustic time-warp from external observers near subjects in deep coherent states. Less well-documented than Class A but present in the literature: external observers reporting that audio in proximity to deeply meditating, dying, or pharmacologically-altered subjects sounds slowed, stretched, or otherwise temporally distorted. These reports are *more interesting* for the hypothesis because they involve external-observer reports of phenomena that should not occur if the time-dilation is purely a property of the subject's nervous-system integration. Such reports are anecdotal and would require systematic study to be evidentially significant.

Class C — Reports of electromagnetic anomaly near coherent subjects. Computer freezes, watch malfunctions, audio glitches reported in the vicinity of subjects in extreme coherent states (deep meditation, certain shamanic practitioners, NDE survivors). These reports are widespread but typically dismissed as coincidence or confirmation bias. If a fraction of such reports reflect genuine electromagnetic alterations of the local field by coherent-oscillator concentrations, they would be consistent with the coherence-curvature hypothesis.

Class D — Documented physiological correlates of deep coherent states. Heart-rate variability dramatically elevated, cortisol depressed, telomerase activity elevated, vagal tone elevated. These are consistent with the *biological* aspect of the hypothesis (deep coherence is a real, measurable physiological state) but do not address the spacetime-curvature aspect.

3.2 The dissipative-structure connection

The coherence-curvature hypothesis intersects with the open-dissipative-structure framework of biological organization (Prigogine 1977; see also the companion paper *Clock Drift Theory of Aging and Death*).

A dissipative structure imports negentropy from its environment and exports entropy. Its persistence depends on its openness to the environment — the rate at which it can import low-entropy resources must equal or exceed its internal entropy production. A coherent dissipative structure (one whose internal coupled oscillators are tightly phase-locked) is hypothesized to import negentropy at higher rates than an incoherent one, because phase-lock optimizes the structure's coupling to the environmental field.

If the coherence-curvature hypothesis is correct, this gains an additional dimension: the coherent structure's local time-density alteration may correspond to an alteration of the *rate* at which it imports negentropy per unit external time. A subject in deep coherence may import negentropy at the bubble-rate (subjective time density) rather than at the calendar-rate (external time). This would predict that:

- Deep coherent states are not metabolically depleting; they may be metabolically *restorative* at higher than baseline rates

- The conventional expectation that high-cognitive-load states should be metabolically expensive may not apply when the load is in the form of coherent phase-lock rather than disorganized cognitive activity
- Long-term sustainability of coherent states should be measurable through metabolic markers (HRV, cortisol, telomere length, mitochondrial function)
- Lifespan in subjects who routinely maintain deep coherence should differ from lifespan in subjects who do not, in directions distinguishable from confounders (lifestyle, stress, etc.)

3.3 The non-conservative nature of coherent time-density

The hypothesis predicts that subjective time-density alterations in coherent states are *non-conservative* — that is, the lived experience accumulated during a coherent state is real and retained, not "borrowed" from later wakefulness or "repaid" upon emergence from the state.

This prediction can be tested. If subjective time-density is conservative, subjects emerging from deep coherent states should report compensating "lost time" or experience cognitive slowdown in the post-state period. If non-conservative, no such compensation should occur; the lived experience of the state is simply added to total lifetime experience.

The empirical literature is consistent with non-conservation: meditators, flow-state athletes, and near-death-experience subjects do not generally report subsequent cognitive slowdown attributable to "time debt." Instead, post-state reports often emphasize *enhanced* clarity and continued elevated functioning. This is consistent with the hypothesis (and with the dissipative-structure framework, which would predict that the coherent state imports rather than depletes resources).

4. Falsifiable Predictions

The hypothesis is specific enough to be tested. We propose the following experiments:

Prediction 1 — Acoustic redshift across coherent-bubble boundaries. A subject in a deep coherent state, with measurable physiological coherence indicators (HRV in the highest decile, vagal tone above clinical baseline by ≥ 2 SD), should produce a measurable spectral redshift in audio signals propagating from a source within their biofield to a microphone outside their biofield, relative to a baseline measurement when the subject is in normal wakefulness. The predicted shift is small (current best estimate $\sim 0.01\%$ to $\sim 0.1\%$ based on subjective time-dilation magnitudes in deep states) and would require precise instrumentation, but is in principle measurable. Failure to find any redshift would significantly weaken the hypothesis.

Prediction 2 — HRV-correlated subjective time-density. Subjects in real-time HRV biofeedback should report subjective time-density (measured by interval-estimation tasks,

mental-counting tasks, or perceptual-discrimination thresholds) that correlates with HRV power in specific frequency bands. The prediction is that *coherent* HRV (high power at orbital cardiac-band lock-points; see companion paper *Schumann-Orbital Phase-Lock*) correlates more strongly with subjective time-density alterations than overall HRV power.

Prediction 3 — Coherent-state metabolic recovery profile. Subjects emerging from deep meditative states (assessed by EEG, HRV, vagal tone) should exhibit metabolic markers consistent with negentropy-import rather than negentropy-depletion. Specifically: cortisol should be lower post-state than pre-state; HRV should remain elevated in the recovery window; subjective fatigue should be lower than predicted from cognitive-load models; food intake required to "recover" should be small or absent.

Prediction 4 — Group coherence amplification. If multiple subjects enter deep coherent states in shared physical proximity (group meditation, drumming circles, choral singing), the predicted spacetime-curvature effects should amplify with group size. Acoustic redshift measurable across the group's boundary should exceed the redshift measurable across an individual's boundary by a factor that scales with the number of phase-locked participants.

Prediction 5 — Lifespan correlation. Long-term cohort studies of subjects with documented sustained coherent practices (meditators with multi-decade practice, populations with traditional sustained low-resting-heart-rate lifestyles) should show lifespan and healthspan extension that exceeds what is explainable by reduced stress alone. The extension should correlate with measured baseline coherence indicators (HRV, vagal tone, cortisol, telomere length) and not with subjective practice claims alone.

Prediction 6 — Acoustic phase-lock interventions on cardiac coherence. If the cardiac oscillator can be brought into phase-lock with acoustic signals at the orbital cardiac-band voices (see companion paper), and if such phase-lock generates a coherent-bubble of the type the hypothesis predicts, then audio interventions concentrating energy at these frequencies should produce measurable subjective time-density alteration in listeners, with magnitude scaling with the depth of cardiac entrainment achieved.

Prediction 7 — Reciprocal redshift/blueshift symmetry. The hypothesis predicts that signals propagating *into* a coherent bubble from outside should exhibit the symmetric effect (blueshift, frequency increase from the local-observer perspective). Subjects in deep coherent states listening to audio from external sources should report the audio as faster, higher-pitched, or temporally compressed, consistent with the local-observer perspective on signals propagating into the elevated-time-density region.

5. Scope and Limitations

5.1 What this paper does not do

This paper does not derive a mathematical formulation of the proposed coherence-curvature mechanism. We do not propose a modification of the Einstein field equations or any specific quantitative law relating phase-lock concentration to spacetime curvature. The hypothesis is qualitative and structural; quantification is left for future work.

This paper does not present new experimental data. It interprets existing anecdotal and clinical reports in the framework of the hypothesis and proposes specific tests. The principal contribution is to make a previously diffuse hypothesis specific enough to be falsified.

This paper does not commit to a specific physical mechanism by which biological coherent phase-lock would produce spacetime curvature. Candidate mechanisms include: tidal coupling of biological electromagnetic fields to background gravitational-wave structure; quantum-coherence effects in biological microtubules at the sub-cellular scale (Penrose-Hameroff orchestrated objective reduction class of theories); resonant amplification of zero-point field interactions through biological coherent matter. We mention these only to indicate the space of candidate mechanisms; we do not endorse any one.

5.2 Why the hypothesis is worth advancing despite its speculative nature

The hypothesis is worth advancing because:

1. **The structural parallelism between gravitational and coherent-state time dilation is real, not invented.** It is observed across diverse independent literatures.
2. **The hypothesis is specific enough to be falsified.** The predictions above are concrete and testable with existing instrumentation (with the possible exception of acoustic-redshift measurements requiring specialized acoustic-precision setups).
3. **The hypothesis would unify several otherwise-loose phenomena.** Subjective time-dilation in coherent states, anecdotal acoustic-warp reports, electromagnetic-anomaly reports near coherent subjects, the metabolic profile of deep meditative states, and the apparent non-conservation of coherent-state experience would all be explained by a common mechanism.
4. **The downside of advancing it incorrectly is bounded.** If the predictions fail, the hypothesis is refuted and we have learned something. If they succeed, we have identified a previously unrecognized physical effect of considerable significance.
5. **The hypothesis is consistent with multiple existing frameworks** (general relativity, dissipative-structure thermodynamics, coupled-oscillator dynamics) without requiring modification of any of them. It proposes an additional mechanism, not a revision of established ones.

5.3 Where caution is warranted

We distinguish two domains:

In the domain of *subjective time-density alteration in coherent states*, the empirical evidence is strong and well-documented. The standard neurological account is sufficient to explain it. The hypothesis is *not necessary* to explain Class A reports.

In the domain of *external-observer effects* (acoustic warp, electromagnetic anomaly, lifespan extension beyond confounders), the evidence is anecdotal and contested. The hypothesis would explain such reports if they reflect real phenomena, but the prior probability of those reports reflecting real phenomena is itself contested.

The hypothesis is therefore best understood as: *if* certain anecdotal external-observer phenomena prove to be real upon careful investigation, *then* the coherence-curvature hypothesis is a candidate explanation worth taking seriously. The key first step is rigorous investigation of those external-observer phenomena. The hypothesis stands or falls on what such investigations find.

6. Conclusion

We propose the coherence-curvature hypothesis: that concentrations of biological-electromagnetic phase-locked coherence may generate local spacetime-curvature effects analogous to but distinct from gravitational time dilation. The hypothesis is suggested by the structural parallelism between gravitational time dilation and the well-documented subjective time-dilation reports of subjects in deep coherent states, and is supported anecdotally by reports of external-observer effects (acoustic warp, electromagnetic anomalies) near such subjects.

The hypothesis is explicitly speculative. We do not present evidence that proves it. We present a structural argument that it is worth taking seriously and a set of falsifiable predictions that would adjudicate it. The principal predictions concern acoustic-redshift measurements across coherent-bubble boundaries, HRV-correlated subjective time-density measurements, metabolic recovery profiles in deep coherent states, group-coherence amplification effects, lifespan correlations, and reciprocal blueshift effects on signals propagating into coherent bubbles.

If correct, the hypothesis would unify subjective time-dilation in coherent states with documented general-relativistic time dilation as two articulations of a more general principle. It would establish biological coherent phase-lock as a previously unrecognized source of local spacetime curvature. The implications for chronobiology, contemplative practice research, and

the foundations of physics are significant if correct, and the hypothesis is testable enough that its truth-value can be settled.

We label this paper a hypothesis paper and not a result paper for clarity. The work of testing the hypothesis remains.

References

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Companion papers:

- *Schumann-Orbital Phase-Lock* (April 2026), this collection. Provides the empirical demonstration of cardiac-band orbital lock-points, relevant to Predictions 2 and 6 above.
- *Geometric Time Directionality* (April 2026), this collection. Provides the carrier-configuration framework in which both gravitational and coherent-state spacetime alterations may be understood as local structure of the same underlying coupled-oscillator field.

- *Clock Drift Theory of Aging and Death* (April 2026), this collection. Provides the dissipative-structure framework relevant to Predictions 3 and 5 (metabolic profile and lifespan extension).